What Have I Learned About Energy?

Unit 1 Study Guide · Energy Transfer & Transformation · Teacher Key

Name: _____ **Date:** _____

A · MATCH THE WORD

1-F · 2-J · 3-D · 4-B · 5-I · 6-A · 7-H · 8-E · 9-C · 10-G

B · FILL IN THE BLANK

1. cause change · 2. evidence · 3. stored · 4. motion · 5. faster · 6. transfer

C · TRANSFER, TRANSFORMATION, OR BOTH?

1. TRANSFER (heat moves, stays heat) · 2. TRANSFORMATION (stored → light) · 3. BOTH (motion passes + sound) · 4. TRANSFORMATION (electric current → motion)

D · SAME OBJECT, DIFFERENT SPEED

cup moves FARTHER · a faster car has MORE energy of motion. MEASURE: ramp height / how far the cup moves. INFER: speed and energy of motion.

E · LABEL THE ENERGY CHANGE

INPUT: stored energy (battery) → OUTPUT: light (and a little heat).

F · WHAT HAPPENS IN A COLLISION?

energy does NOT disappear · transfers to the target (some becomes sound).

G · ENGINEERING WORDS

Criteria = what a good solution must do · Constraints = the limits (materials, time, safety) · Refine = to improve using test evidence.

H · EDISON & LATIMER

improved · 6,000 · bamboo · filament.

I · SHORT RESPONSE

Any device: claim (what it does) + evidence (what is observed) + reasoning (why the evidence supports the claim), using “supports.” Example: flashlight — claim: a flashlight changes stored energy into light; evidence: the room gets brighter when it turns on; reasoning: the brightness is the change that supports the claim.